Step 2: Buggy Application Output

```
$ python3 tennis_scoring.py
==== Running parse_score ====
Enter points won as A-B (e.g. 3-2): 3-2
Points A and B: (3, 2)

==== Running game_lead ====
Enter a game score (e.g. 40-30): 40-30
Lead in Points: 1

==== Running set_spread ====
Enter games won per set as CSV (e.g. 6,4,7): 6,4,7
Spread: 3
```

```
$ pytest test_tennis_scoring.py
============================= test session starts ==============================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7
plugins: hypothesis-6.141.1
collected 3 items

test_tennis_scoring.py FFF                                               [100%]

=================================== FAILURES ===================================
_____________________________ test_parse_score_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 5, in test_parse_score_no_crash
  |     def test_parse_score_no_crash(s):
  |   File "/Users/shyam/Library/Python/3.9/lib/python/site-packages/hypothesis/core.py", line 2124, in wrapped_test
  |     raise the_error_hypothesis_found
  | exceptiongroup.ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 7, in test_parse_score_no_crash
    |     parse_score(s)
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/tennis_scoring.py", line 3, in parse_score
    |     return int(a), int(b)
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 9, in test_parse_score_no_crash
    |     assert False, f"parse_score crashed with: {e}"
    | AssertionError: parse_score crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Falsifying example: test_parse_score_no_crash(
    |     s='-',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 7, in test_parse_score_no_crash
    |     parse_score(s)
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/tennis_scoring.py", line 2, in parse_score
    |     a, b = text.split('-')
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 9, in test_parse_score_no_crash
    |     assert False, f"parse_score crashed with: {e}"
    | AssertionError: parse_score crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Falsifying example: test_parse_score_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 7, in test_parse_score_no_crash
    |     parse_score(s)
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/tennis_scoring.py", line 2, in parse_score
    |     a, b = text.split('-')
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 9, in test_parse_score_no_crash
    |     assert False, f"parse_score crashed with: {e}"
    | AssertionError: parse_score crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Falsifying example: test_parse_score_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________ test_game_lead_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 12, in test_game_lead_no_crash
  |     def test_game_lead_no_crash(s):
  |   File "/Users/shyam/Library/Python/3.9/lib/python/site-packages/hypothesis/core.py", line 2124, in wrapped_test
  |     raise the_error_hypothesis_found
  | exceptiongroup.ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 14, in test_game_lead_no_crash
    |     game_lead(s)
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/tennis_scoring.py", line 8, in game_lead
    |     return points[first] - points[second]
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 16, in test_game_lead_no_crash
    |     assert False, f"game_lead crashed with: {e}"
    | AssertionError: game_lead crashed with: ''
    | assert False
    | Falsifying example: test_game_lead_no_crash(
    |     s='-',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 14, in test_game_lead_no_crash
    |     game_lead(s)
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/tennis_scoring.py", line 7, in game_lead
    |     first, second = text.split('-')
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 16, in test_game_lead_no_crash
    |     assert False, f"game_lead crashed with: {e}"
    | AssertionError: game_lead crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Falsifying example: test_game_lead_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 14, in test_game_lead_no_crash
    |     game_lead(s)
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/tennis_scoring.py", line 7, in game_lead
    |     first, second = text.split('-')
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 16, in test_game_lead_no_crash
    |     assert False, f"game_lead crashed with: {e}"
    | AssertionError: game_lead crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Falsifying example: test_game_lead_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________ test_set_spread_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 19, in test_set_spread_no_crash
  |     def test_set_spread_no_crash(s):
  |   File "/Users/shyam/Library/Python/3.9/lib/python/site-packages/hypothesis/core.py", line 2124, in wrapped_test
  |     raise the_error_hypothesis_found
  | exceptiongroup.ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 21, in test_set_spread_no_crash
    |     set_spread(s)
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/tennis_scoring.py", line 11, in set_spread
    |     games = [int(g) for g in games_csv.split(',')]
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/tennis_scoring.py", line 11, in <listcomp>
    |     games = [int(g) for g in games_csv.split(',')]
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 23, in test_set_spread_no_crash
    |     assert False, f"set_spread crashed with: {e}"
    | AssertionError: set_spread crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Falsifying example: test_set_spread_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 21, in test_set_spread_no_crash
    |     set_spread(s)
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/tennis_scoring.py", line 11, in set_spread
    |     games = [int(g) for g in games_csv.split(',')]
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7/test_tennis_scoring.py", line 23, in test_set_spread_no_crash
    |     assert False, f"set_spread crashed with: {e}"
    | AssertionError: set_spread crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Falsifying example: test_set_spread_no_crash(
    |     s=None,
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_tennis_scoring.py::test_parse_score_no_crash - exceptiongroup.Exc...
FAILED test_tennis_scoring.py::test_game_lead_no_crash - exceptiongroup.Excep...
FAILED test_tennis_scoring.py::test_set_spread_no_crash - exceptiongroup.Exce...
============================== 3 failed in 0.26s ===============================
```

Step 5: Fixed Application Output

```
$ python3 tennis_scoring.py
==== Running parse_score ====
Enter points won as A-B (e.g. 3-2): 3-2
Points A and B: (3, 2)

==== Running game_lead ====
Enter a game score (e.g. 40-30): 40-30
Lead in Points: 1

==== Running set_spread ====
Enter games won per set as CSV (e.g. 6,4,7): 6,4,7
Spread: 3
```

Step 6: Hypothesis Tests Passed

```
$ pytest test_tennis_scoring.py
============================= test session starts ==============================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/shyam/Documents/Lab6&7SE/L6_L7_34_tennis_scoring/lab7
plugins: hypothesis-6.141.1
collected 3 items

test_tennis_scoring.py ...                                               [100%]

============================== 3 passed in 0.32s ===============================
```